

# **Boosting Regression Algorithms in Machine Learning**

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# Why Do We Need Boosting?

## Challenges in Regression

- Linear Regression struggles with complex nonlinear patterns
- Single Decision Trees often overfit
- Ensemble methods improve generalization
- Boosting significantly improves prediction accuracy

## Traditional Flow

Dataset → Weak Model → High Error

## Boosting Flow

Dataset → Weak Model 1 → Weak Model 2 → Weak Model 3 → Strong Model

## Key Benefits

- ✓ Better Accuracy
- ✓ Handles Nonlinearity
- ✓ Feature Interactions
- ✓ State-of-the-Art Performance

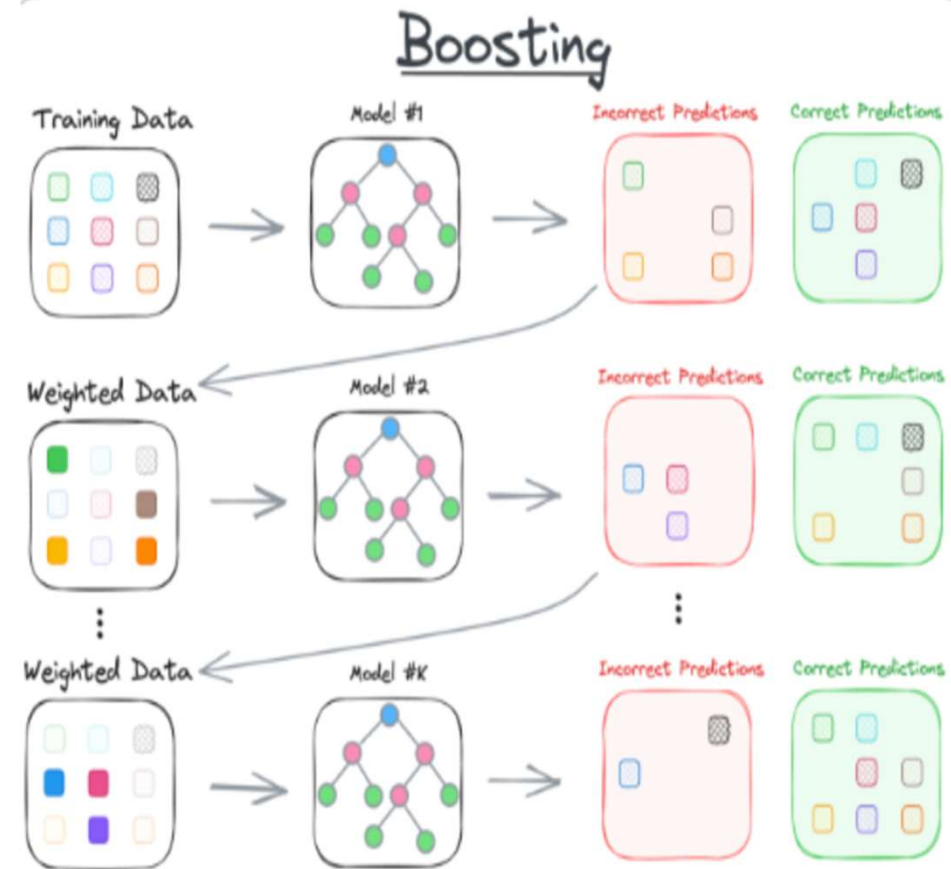
# What is Boosting?

## Definition

- Boosting is an ensemble learning technique where models are trained sequentially.
- Each new model focuses on correcting the errors (residuals) made by the previous model.

## Characteristics

- Sequential learning
- Weak learners become strong learners
- Primarily uses shallow decision trees
- Improves predictive power



# Types of Boosting Algorithms

## Boosting Family

- AdaBoost
- Gradient Boosting
- HistGradient Boosting
- XGBoost
- LightGBM
- CatBoost

### Evolution

AdaBoost → Gradient Boosting → HistGradient Boosting → XGBoost →  
LightGBM → CatBoost

| Comparison            |                                      |            |           |           |
|-----------------------|--------------------------------------|------------|-----------|-----------|
| Algorithm             | Key Innovation                       | Complexity | Speed     | Accuracy  |
| AdaBoost              | Sample re-weighting                  | Low        | Medium    | Good      |
| Gradient Boosting     | Residual error learning              | Medium     | Slow      | High      |
| HistGradient Boosting | Histogram-based split finding        | Medium     | Fast      | High      |
| XGBoost               | Regularization + parallel processing | High       | Fast      | Very High |
| LightGBM              | Leaf-wise growth, faster training    | High       | Very Fast | High      |
| CatBoost              | Native categorical feature handling  | High       | Fast      | High      |

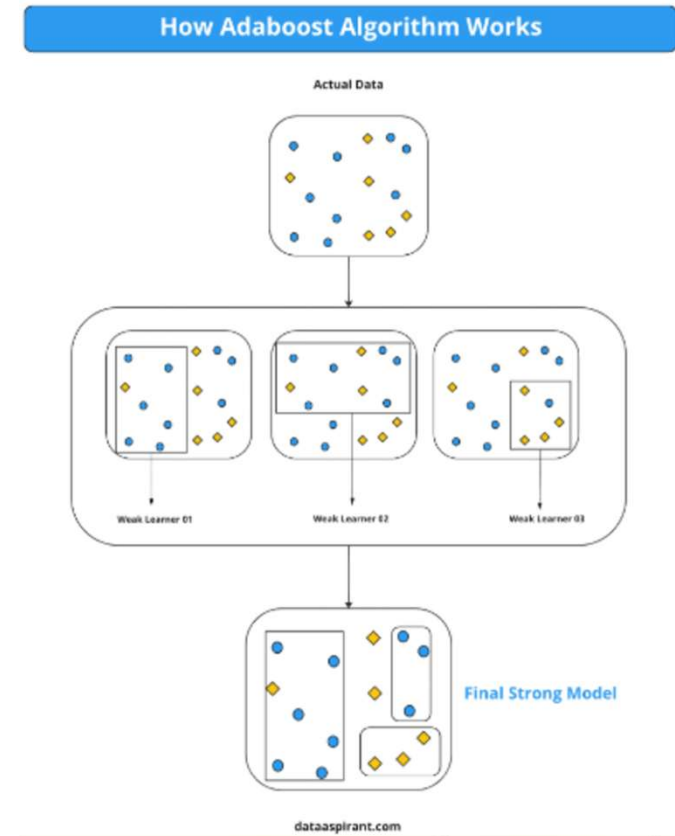
# 1. AdaBoost Regression

## Concept

- Adaptive Boosting assigns higher weights to poorly predicted records.
- Future learners focus more on difficult observations.

## Workflow

- Train first tree
- Compute errors
- Increase weights of difficult samples
- Train next tree
- Repeat
- Combine outputs



# AdaBoost Python Implementation

```
1 from sklearn.ensemble import AdaBoostRegressor
2 from sklearn.tree import DecisionTreeRegressor
3
4 model = AdaBoostRegressor(
5     estimator=DecisionTreeRegressor(max_depth=3),
6     n_estimators=100,
7     learning_rate=0.1,
8     random_state=42
9 )
10
11 model.fit(X_train, y_train)
12
13 y_pred = model.predict(X_test)
```

| Important Parameters |                 |
|----------------------|-----------------|
| Parameter            | Purpose         |
| estimator            | Base learner    |
| n_estimators         | Number of trees |
| learning_rate        | Learning speed  |
| loss                 | Error weighting |

# 2.Gradient Boosting Regression

## Main Idea

Rather than reweighting data points:

Gradient Boosting learns residual errors.

## Mathematical Concept

### Residual:

$r = \text{Actual} - \text{Prediction}$

New Tree learns:

$r_1$

$r_2$

$r_3$

...

## Update

- $F_m(x) = F_{m-1}(x) + \eta h_m(x)$
- Where:
- $\eta$  = Learning Rate
- $h_m$  = New Tree

## Steps

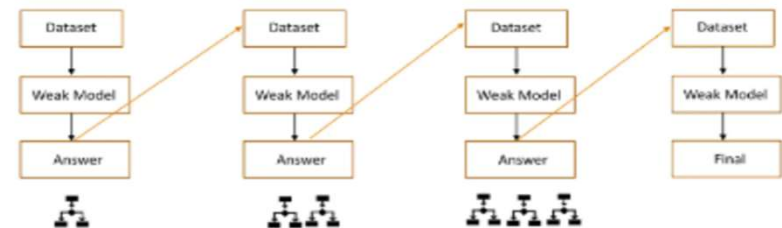
- Predict Mean Value
- Calculate Residuals
- Train Tree on Residuals
- Add New Prediction
- Repeat

## Advantages

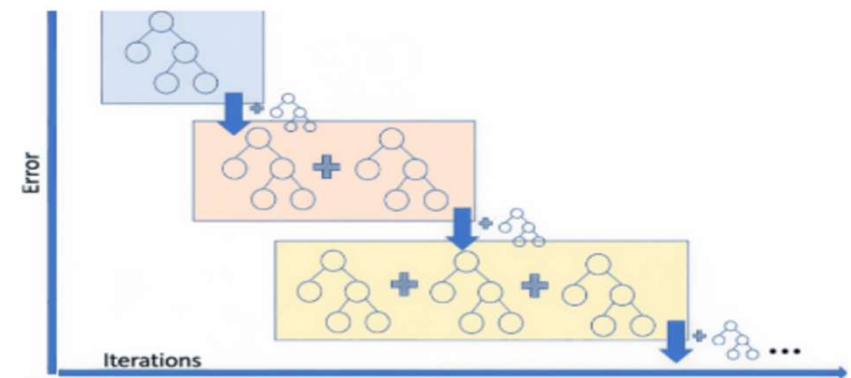
- High Accuracy
- Flexible Loss Functions
- Handles Nonlinearity

## What is Gradient Boosting

Gradient boosting is a technique for building an ensemble of weak models such that the predictions of the ensemble minimize a loss function. We are combining the predictions of multiple models, so we are not optimizing the model parameters directly but the boosted model predictions, therefore, the gradients.



## Gradient Boosting Regressor



Source :  datacamp

# Gradient Boosting Regressor Python Implementation

```
1 from sklearn.ensemble import GradientBoostingRegressor
2
3 gbr = GradientBoostingRegressor(
4     n_estimators=200,
5     learning_rate=0.05,
6     max_depth=4
7 )
8
9 gbr.fit(X_train,y_train)
```

## Important Hyperparameters

- learning\_rate
- n\_estimators
- max\_depth
- subsample

# 3.HistGradient Boosting Regressor

## Why It Was Introduced?

HistGradientBoostingRegressor is much faster for datasets having more than 10,000 samples and supports native missing value handling.

## Advantages

-  Faster Training
-  Missing Value Support
-  Inspired by LightGBM
-  Lower Memory Usage

```
1 from sklearn.ensemble import HistGradientBoostingRegressor
2
3 model = HistGradientBoostingRegressor(
4     learning_rate=0.05,
5     max_iter=300)
6
7 model.fit(X_train,y_train)
```

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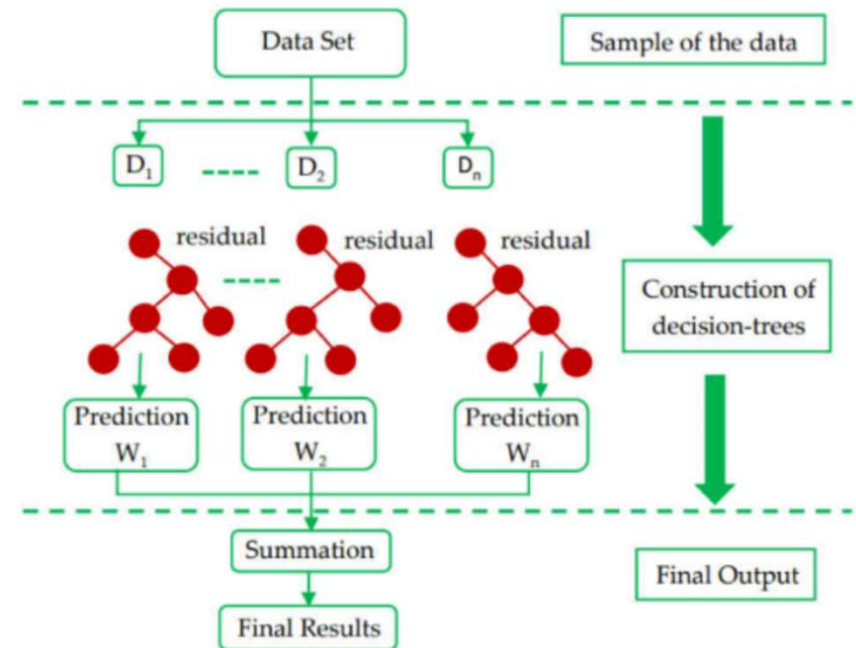
# 4.XGBoost

## Extreme Gradient Boosting

Enhancement of Gradient Boosting with:

- Regularization
- Parallel Processing
- Tree Pruning
- Optimized Split Finding

```
1 from xgboost import XGBRegressor
2
3 model = XGBRegressor(
4     n_estimators=500,
5     max_depth=6,
6     learning_rate=0.05)
7
8 model.fit(X_train,y_train)
```



XGBoost (extreme gradient boosting) algorithm structure [31].

# 5.LightGBM

## Light Gradient Boosting Machine

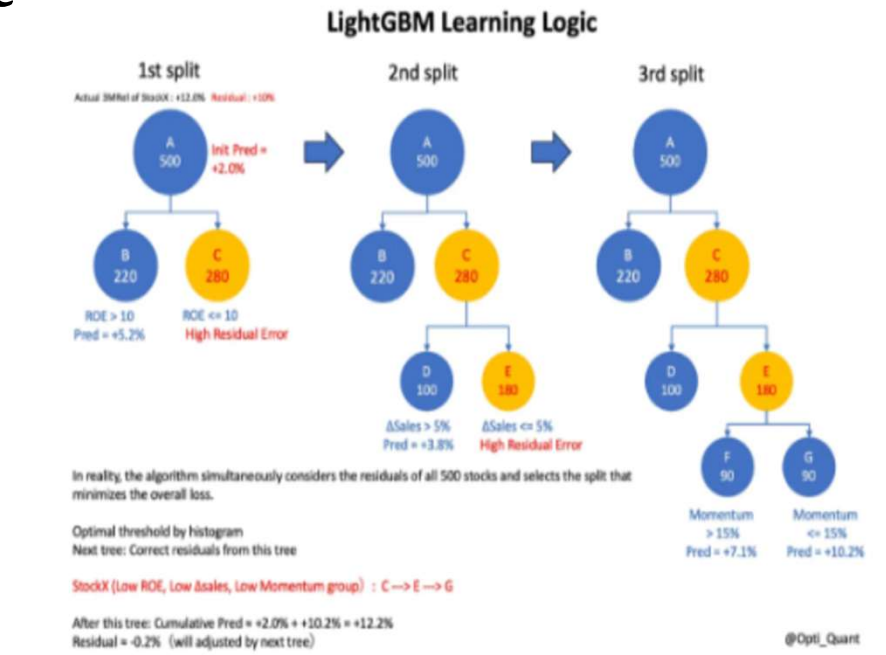
### Key Innovation

- Leaf-wise Tree Growth

### Advantages:

- Faster Training
- Lower Memory
- Large Dataset Support

```
1 import lightgbm as lgb
2
3 model = lgb.LGBMRegressor(
4     n_estimators=500,
5     learning_rate=0.05)
6
7 model.fit(X_train,y_train)
```



# 6.CatBoost

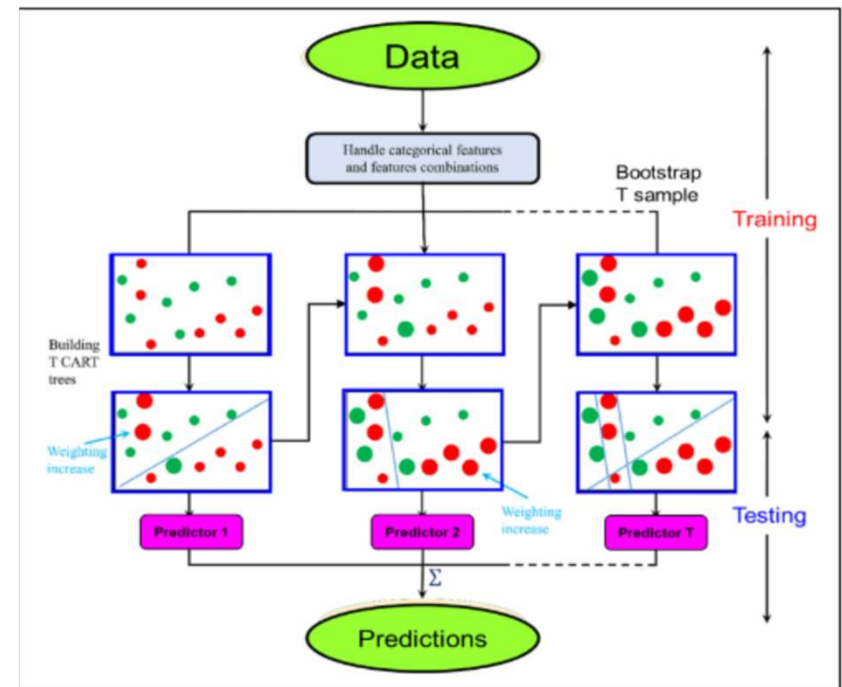
## Categorical Boosting

Designed by Yandex

### Main Innovations

- Ordered Boosting
- Oblivious Trees
- Native Categorical Handling

```
1 from catboost import CatBoostRegressor
2
3 model = CatBoostRegressor(
4     iterations=500,
5     depth=6,
6     learning_rate=0.05)
7
8 model.fit(X_train,y_train)
```



# Final Comparison

| Algorithm            | Speed            | Accuracy         | Best For                  |
|----------------------|------------------|------------------|---------------------------|
| AdaBoost             | <i>Medium</i>    | <i>Good</i>      | <i>Small Data</i>         |
| Gradient Boosting    | <i>Slow</i>      | <i>High</i>      | <i>General Regression</i> |
| HistGradientBoosting | <i>Fast</i>      | <i>High</i>      | <i>Large Data</i>         |
| XGBoost              | <i>Fast</i>      | <i>Very High</i> | <i>Competitions</i>       |
| LightGBM             | <i>Very Fast</i> | <i>High</i>      | <i>Big Data</i>           |
| CatBoost             | <i>Fast</i>      | <i>High</i>      | <i>Categorical Data</i>   |